

Lecture 10

EQUATIONS OF MOTION: CYLINDRICAL COORDINATES

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EQUATIONS OF MOTION: CYLINDRICAL COORDINATES

Today's Objectives:

Students will be able to:

1. Analyze the kinetics of a particle using cylindrical coordinates.



In-Class Activities:

- Reading Quiz
- Applications
- Equations of Motion using Cylindrical Coordinates
- Angle between Radial and Tangential Directions
- Example Problem
- Group Problem Solving
- Attention Quiz

READING QUIZ

1. The normal force which the path exerts on a particle is always perpendicular to the _____.
A) radial line. B) transverse direction.
C) tangent to the path. D) None of the above.
2. When the forces acting on a particle are resolved into cylindrical components, friction forces always act in the _____ direction.
A) radial B) tangential
C) transverse D) None of the above.

APPLICATIONS



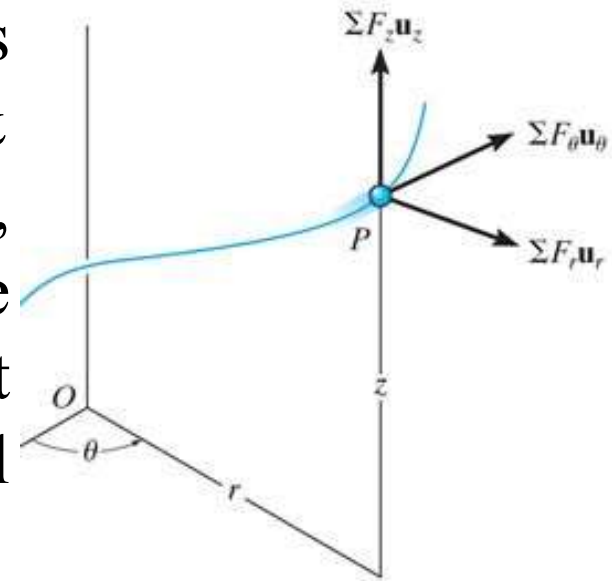
The forces acting on the 50-kg boy can be analyzed using the cylindrical coordinate system.

How would you write the equation describing the frictional force on the boy as he slides down this helical slide?

CYLINDRICAL COORDINATES

(Section 13.6)

This approach to solving problems has some **external similarity** to the normal & tangential method just studied. However, the path may be more complex or the problem may have other attributes that make it desirable to use cylindrical coordinates.



Equilibrium equations or “Equations of Motion” in cylindrical coordinates (using r , θ , and z coordinates) may be expressed in scalar form as:

$$\begin{aligned}\Sigma F_r &= ma_r = m (r - r \dot{\theta}^2) & \dots & \cdot \\ \Sigma F_\theta &= ma_\theta = m (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) & \dots & \cdot \cdot \\ \Sigma F_z &= ma_z = m \ddot{z} & \dots & \end{aligned}$$

CYLINDRICAL COORDINATES (continued)

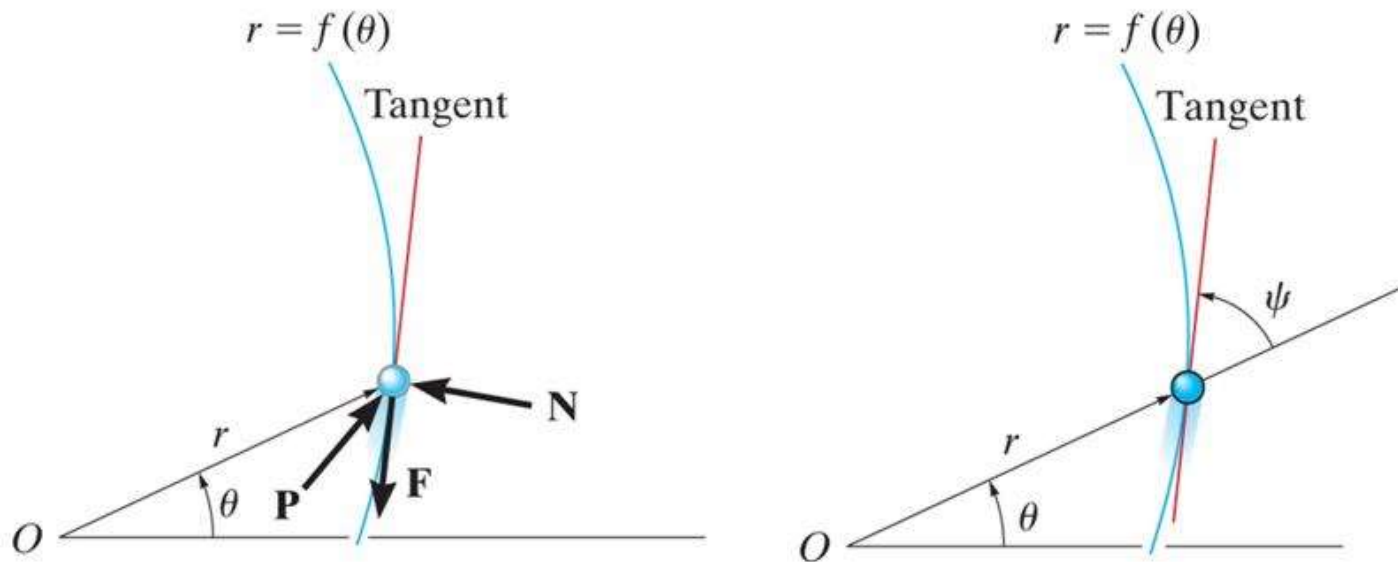
If the particle is constrained to move only in the $r - \theta$ plane (i.e., the z coordinate is constant), then only the first two equations are used (as shown below). The coordinate system in such a case becomes a polar coordinate system. In this case, the path is only a function of θ .

$$\begin{aligned} \sum F_r &= ma_r = m(\ddot{r} - r\dot{\theta}^2) & \ddot{} & \quad \dot{} \\ \sum F_\theta &= ma_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) & \ddot{} & \quad \dot{} \end{aligned}$$

Note that a **fixed** coordinate system is used, not a “body-centered” system as used in the $n - t$ approach.

TANGENTIAL AND NORMAL FORCES

If a force $\mathbf{P}$ causes the particle to move along a path defined by $r = f(\theta)$, the normal force $\mathbf{N}$ exerted by the path on the particle is always perpendicular to the path's tangent. The frictional force $\mathbf{F}$ always acts along the tangent in the opposite direction of motion. The directions of $\mathbf{N}$ and $\mathbf{F}$ can be specified relative to the radial coordinate by using angle ψ .

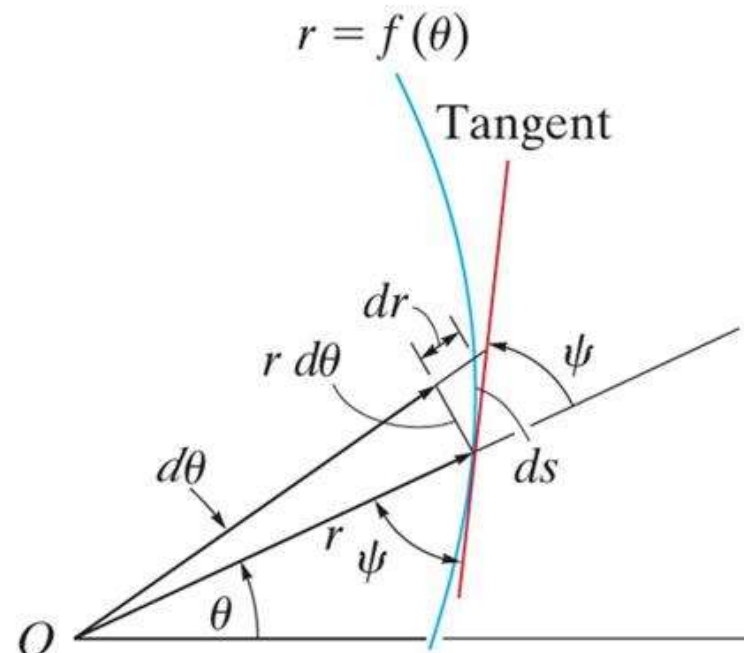


DETERMINATION OF ANGLE ψ

The angle ψ , defined as the angle between the extended radial line and the tangent to the curve, can be required to solve some problems. It can be determined from the following relationship.

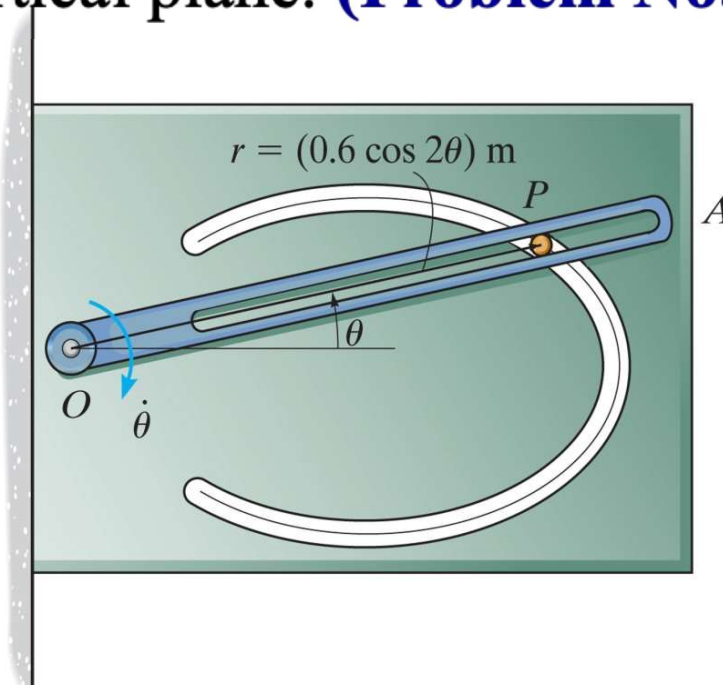
$$\tan \psi = \frac{r d\theta}{dr} = \frac{r}{dr/d\theta}$$

If ψ is positive, it is measured counterclockwise from the radial line to the tangent. If it is negative, it is measured clockwise.

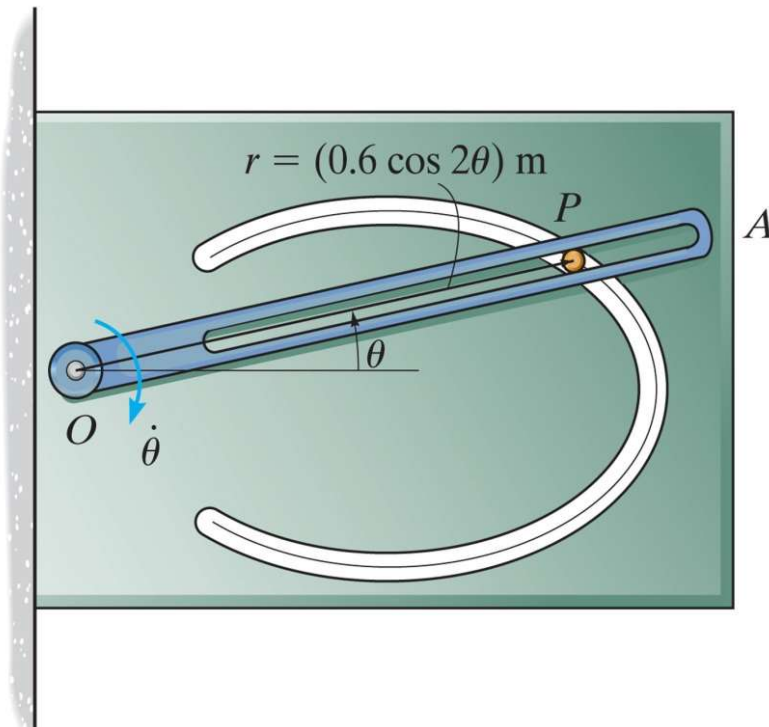


EXAMPLE

The 0.2-kg pin P is constrained to move in the smooth curved slot, which is defined by the lemniscate $r = (0.6 \cos 2\theta)$ m. Its motion is controlled by the rotation of the slotted arm OA , which has a constant clockwise angular velocity of $\dot{\theta} = -3$ rad/s. Determine the force arm OA exerts on the pin P when $\theta = 0^\circ$. Motion is in the vertical plane. **(Problem No. F13-16)**



EXAMPLE (continued)



Given: The 0.2 kg pin (P) is constrained to move in the smooth curved slot, defined by $r = (0.6 \cos 2\theta)$ m. The slotted arm OA has a constant angular velocity of $\dot{\theta} = -3$ rad/s. Motion is in the vertical plane.

Find: Force of the arm OA on the pin P when $\theta = 0^\circ$.

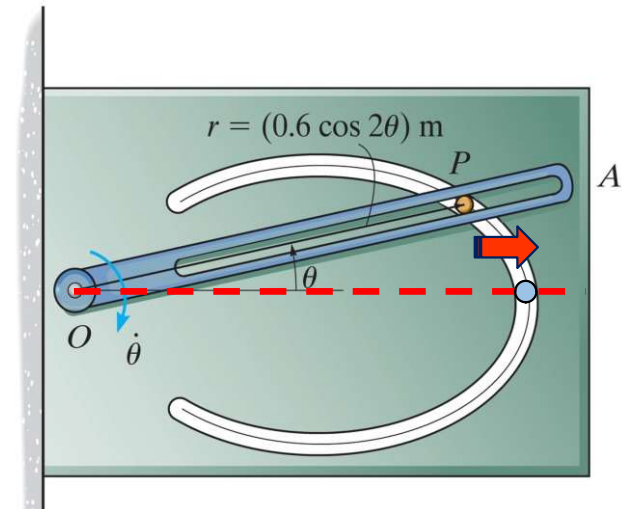
- Plan:**
- 1) Draw the FBD and kinetic diagrams.
 - 2) Develop the kinematic equations using cylindrical coordinates.
 - 3) Apply the equation of motion to find the force.

EXAMPLE (continued)

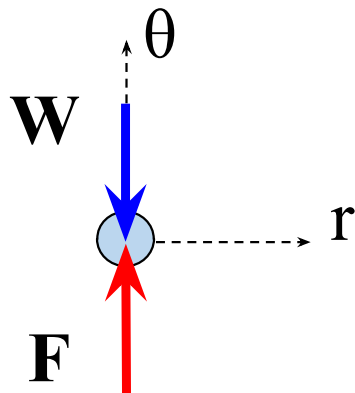
Solution :

1) Free Body and Kinetic Diagrams:

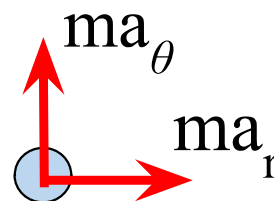
Establish the r, θ coordinate system when $\theta = 0^\circ$, and draw the free body and kinetic diagrams.



Free-body diagram



Kinetic diagram



EXAMPLE (continued)

2) Notice that $r = 0.6 \cos(2\theta)$, therefore:

$$\dot{r} = -1.2 \sin(2\theta) \dot{\theta}$$

$$\ddot{r} = -2.4 \cos(2\theta) \dot{\theta}^2 - 1.2 \sin(2\theta) \ddot{\theta}$$

Kinematics: at $\theta = 0^\circ$, $\dot{\theta} = -3 \text{ rad/s}$, $\ddot{\theta} = 0 \text{ rad/s}^2$.

$$r = 0.6 \cos(0) = 0.6 \text{ m}$$

$$\dot{r} = -1.2 \sin(0) (-3) = 0 \text{ m/s}$$

$$\ddot{r} = -2.4 \cos(0) (-3)^2 - 1.2 \sin(0)(0) = -21.6 \text{ m/s}^2$$

Acceleration components are

$$a_r = \ddot{r} - r\dot{\theta}^2 = -21.6 - (0.6)(-3)^2 = -27 \text{ m/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = (0.6)(0) + 2(0)(-3) = 0 \text{ m/s}^2$$

EXAMPLE (continued)

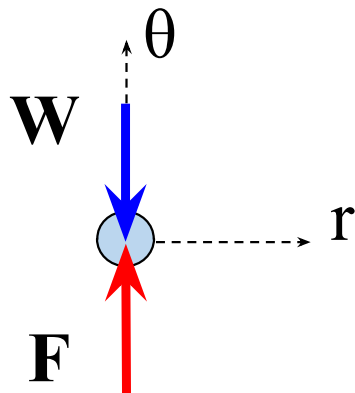
3) Equation of motion: θ direction

$$(+\uparrow) \sum F_{\theta} = ma_{\theta}$$

$$F - 0.2 (9.81) = 0.2 (0)$$

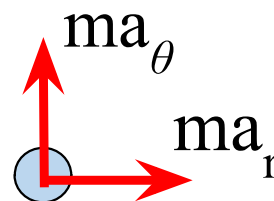
$$F = \underline{1.96 \text{ N}} \uparrow$$

Free-body diagram



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Kinetic diagram

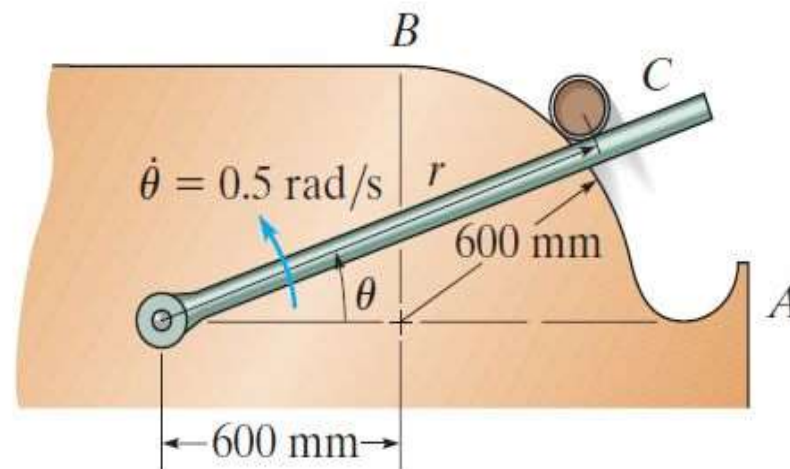


$$a_r = -27 \text{ m/s}^2$$

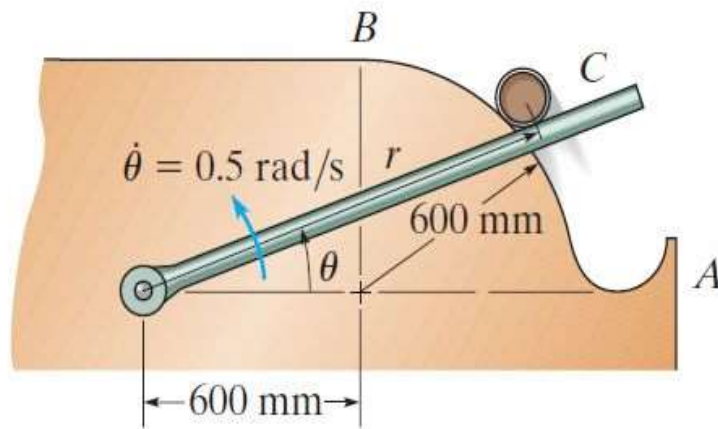
$$a_{\theta} = 0 \text{ m/s}^2$$

GROUP PROBLEM SOLVING

A smooth can C, having a mass of 3 kg, is lifted from a feed at A to a ramp at B by a rotating rod. If the rod maintains a constant angular velocity of $\dot{\theta} = 0.5 \text{ rad/s}$, determine the force which the rod exerts on the can at the instant $\theta = 30^\circ$. Neglect the effects of friction in the calculation and the size of the can so that $r = (1.2 \cos \theta) \text{ m}$. The ramp from A to B is circular, having a radius of 600 mm. **(Problem No. F13-95)**



GROUP PROBLEM SOLVING (continued)



Given: The smooth can C is lifted from A to B by a rotating rod. The mass of can is 3 kg. Neglect the effects of friction in the calculation and the size of the can so that $r = (1.2 \cos \theta) \text{ m}$.

Find: Forces of the rod on the can when $\theta = 30^\circ$ and $\dot{\theta} = 0.5 \text{ rad/s}$, which is constant.

Plan: 1) Find the acceleration components using the kinematic equations.
2) Draw free body diagram & kinetic diagram.
3) Apply the equation of motion to find the forces.

GROUP PROBLEM SOLVING (continued)

Solution:

1) Kinematics:

$$r = 1.2 (\cos \theta)$$

$$\dot{r} = -1.2 (\sin \theta) \dot{\theta}$$

$$\ddot{r} = -1.2 (\cos \theta) \dot{\theta}^2 - 1.2 (\sin \theta) \ddot{\theta}$$

When $\theta = 30^\circ$, $\dot{\theta} = 0.5 \text{ rad/s}$ and $\ddot{\theta} = 0 \text{ rad/s}^2$.

$$r = 1.039 \text{ m}$$

$$\dot{r} = -0.3 \text{ m/s}$$

$$\ddot{r} = -0.2598 \text{ m/s}^2$$

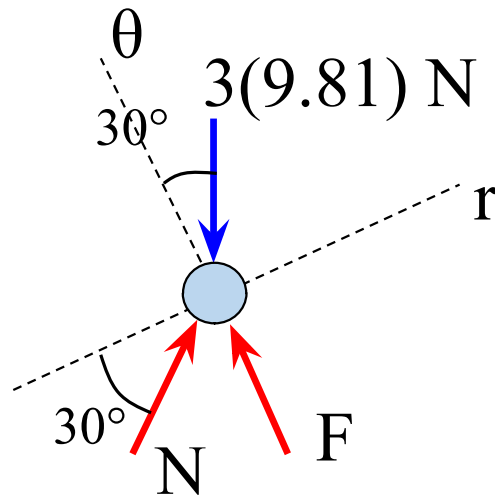
Accelerations:

$$a_r = \ddot{r} - r \dot{\theta}^2 = -0.2598 - (1.039) 0.5^2 = -0.5196 \text{ m/s}^2$$

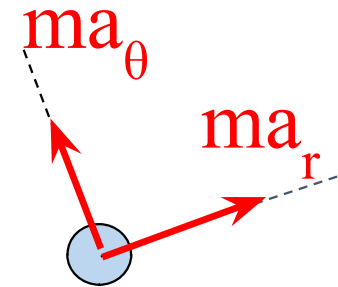
$$a_\theta = r \ddot{\theta} + 2 \dot{r} \dot{\theta} = (1.039) 0 + 2 (-0.3) 0.5 = -0.3 \text{ m/s}^2$$

GROUP PROBLEM SOLVING (continued)

2) Free Body Diagram



Kinetic Diagram



3) Apply equation of motion:

$$\sum F_r = ma_r \Rightarrow -3(9.81) \sin 30^\circ + N \cos 30^\circ = 3(-0.5196)$$

$$\sum F_\theta = ma_\theta \Rightarrow F + N \sin 30^\circ - 3(9.81) \cos 30^\circ = 3(-0.3)$$

$$N = \underline{15.2 \text{ N}}, \quad F = \underline{17.0 \text{ N}}$$

ATTENTION QUIZ

1. For the path defined by $r = \theta^2$, the angle ψ at $\theta = 0.5$ rad is
 - A) 10°
 - B) 14°
 - C) 26°
 - D) 75°
2. If $r = \theta^2$ and $\theta = 2t$, find the magnitude of $\dot{r}$ and $\ddot{\theta}$ when $t = 2$ seconds.
 - A) 4 cm/sec, 2 rad/sec²
 - B) 4 cm/sec, 0 rad/sec²
 - C) 8 cm/sec, 16 rad/sec²
 - D) 16 cm/sec, 0 rad/sec²

End of the Lecture
Let The Learning Continue